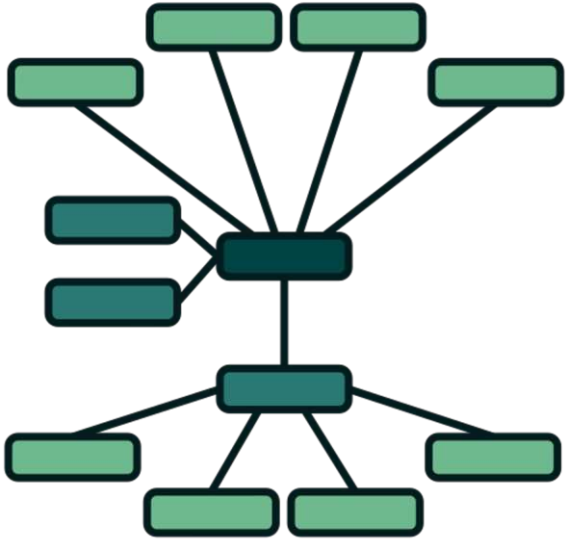


Enriching Ontologies with Disjointness Axioms using Large Language Models

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International Semantic Web Summer School 2024

Problem/Introduction



Ontology





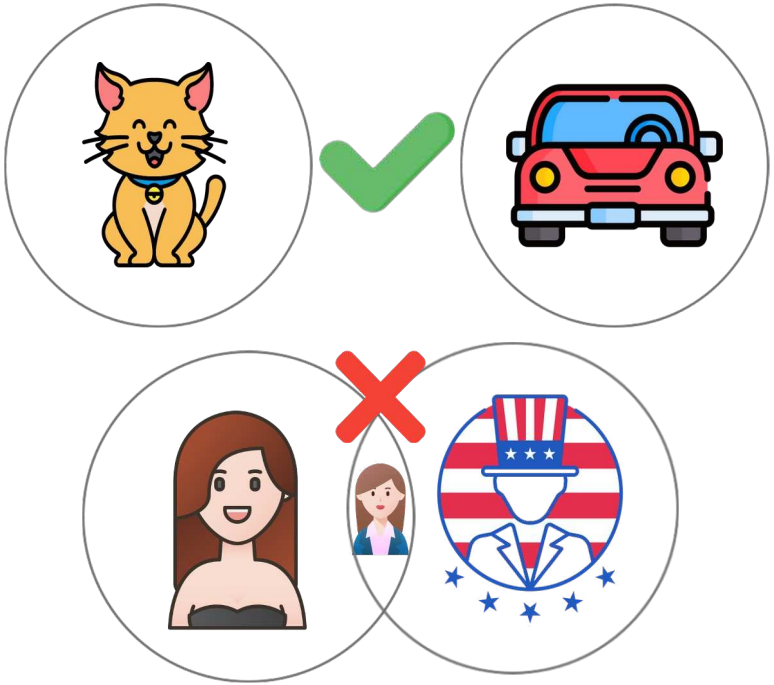
LLM

- Manual curation of Ontologies
- Use LLMs?

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Disjointness

Two classes are considered ”disjoint” if and only if their taxonomic overlap is empty.



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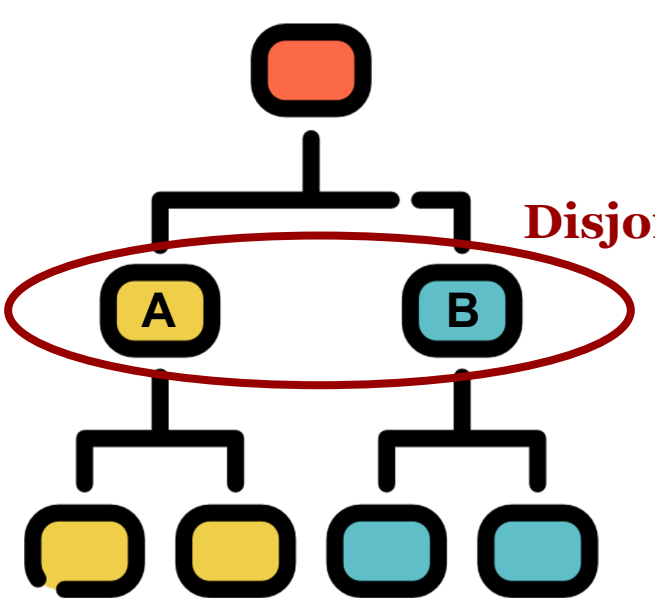
Research Questions

Q1: Can LLMs help enrich ontologies with class disjointness axioms?

Q2: Which LLM prompts work better for disjointness discovery?

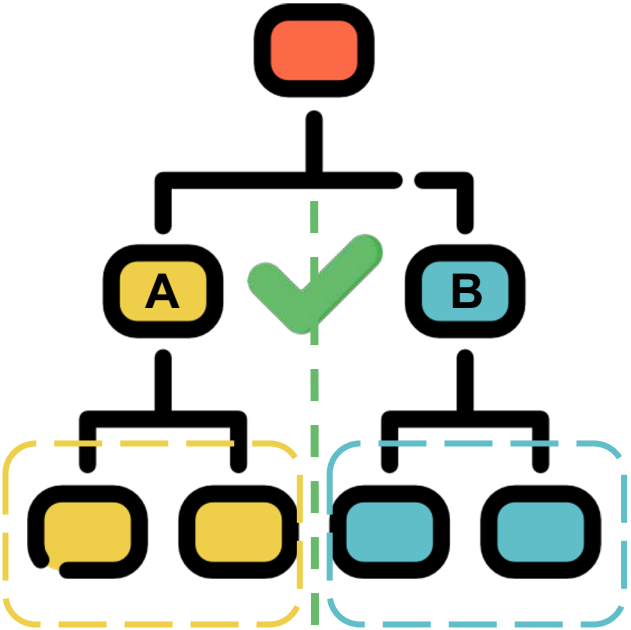
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Approach



Disjoint?

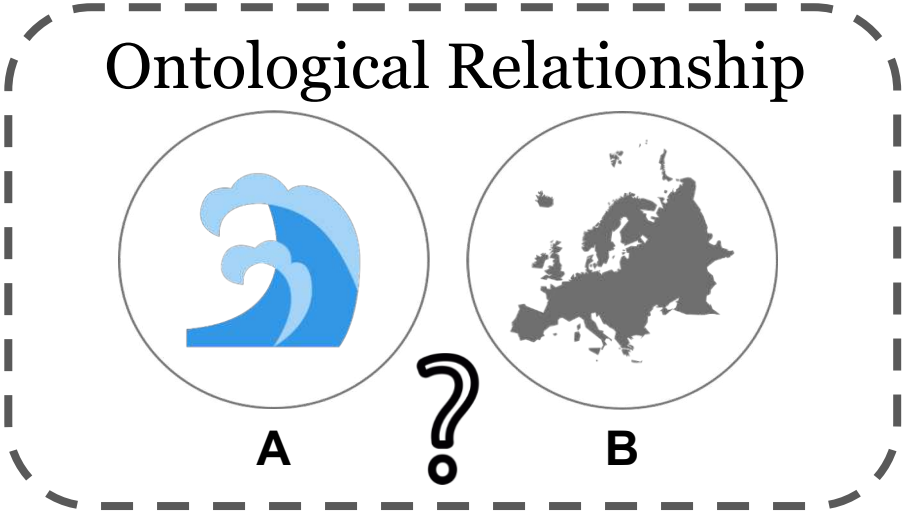
Class Relationship Determination

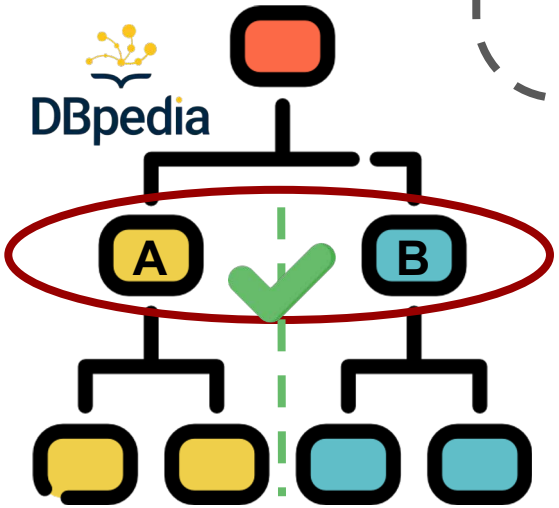


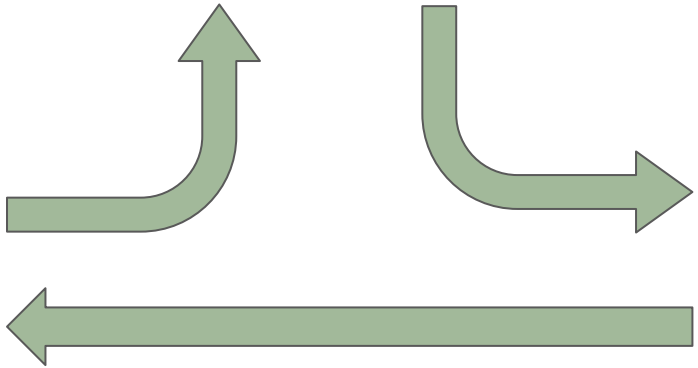
Subclasses **must** be disjoint

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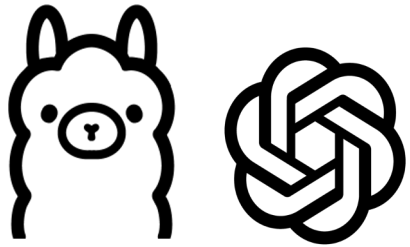
Methodology







LLMs



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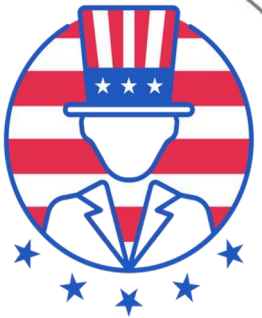
Class Relationships

Negative



Beauty queen





US president

Positive



Sea



Continent

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Prompt strategies

- **Zero-shot naive**

  `Can a "Place" be a "Person"? Answer only with “yes” or “no”.`

- **Zero-shot + Task description**

  `This is a question about ontological disjointness, answer only with “yes” or “no”. Are the classes "Continent" and "Sea" ontologically disjoint?`

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Prompt strategies

- **One-shot + Task description**

  ```
  Example of ontological disjointness is Q: Are the classes "Cat" and "Car" ontologically disjoint? A: Yes. """ Please answer the following question only with “yes” or “no”. Q: Are the classes "Basketball Player" and "Baseball Player" ontologically disjoint?
  ```

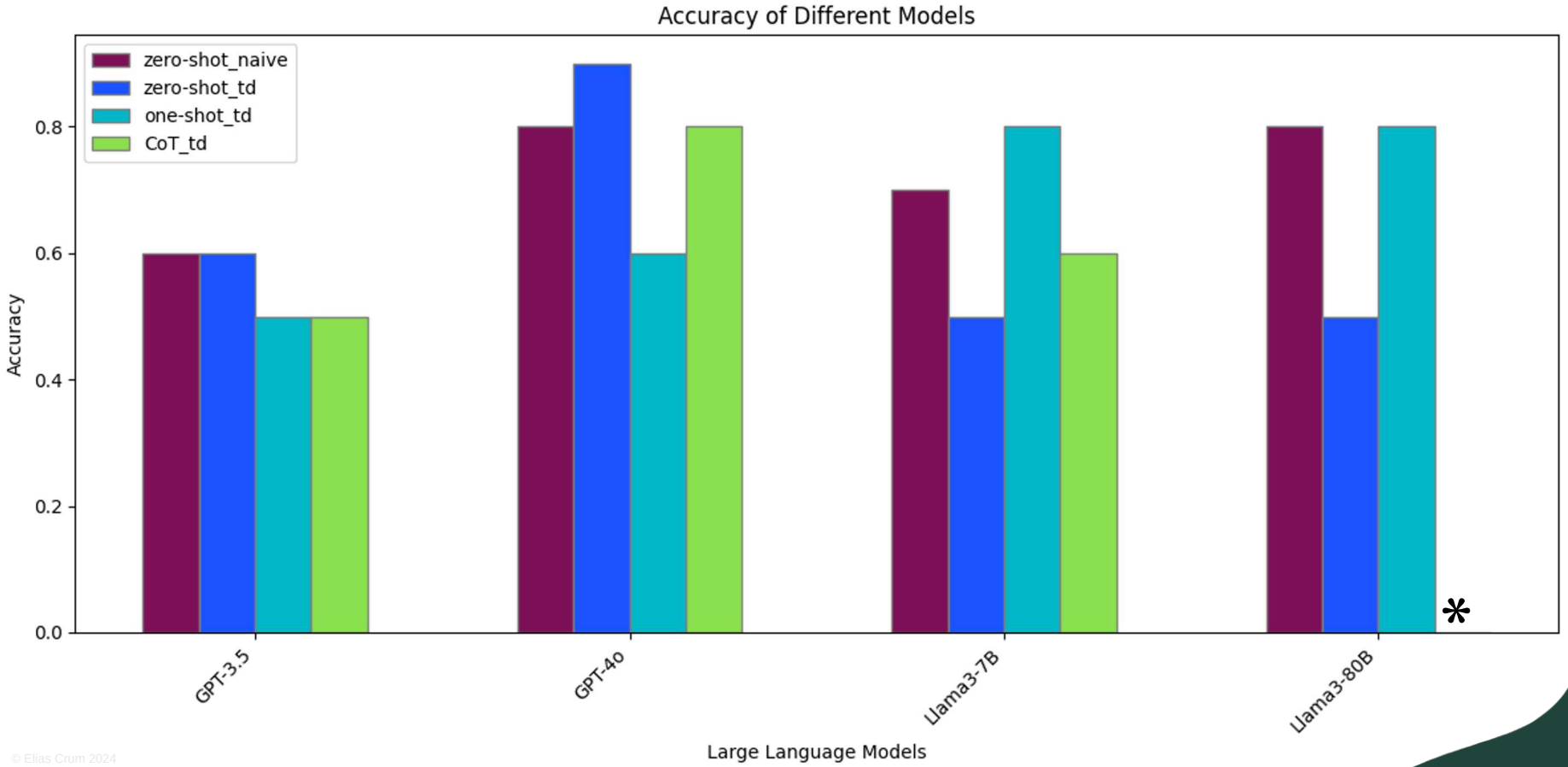
- **Chain-of-Thought + Task description**

  ```
  This is a question about ontological disjointness, final answer only can be “yes” or “no”. Are the classes "Castle" and "Prison" ontologically disjoint? Also, include your reasoning step-by-step.
  ```

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Results:

* = no CoT prompt tested

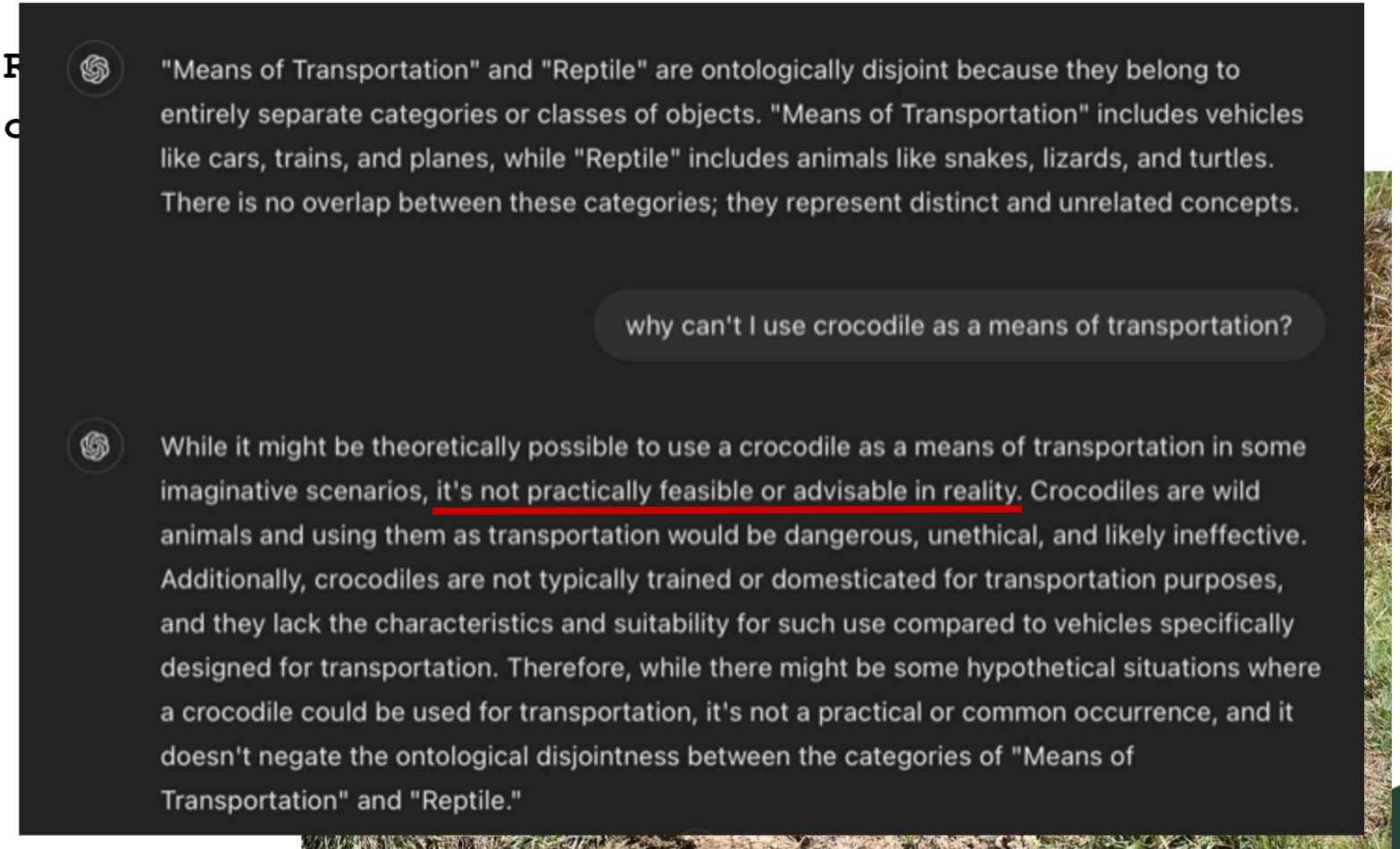


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Limitations

Can a "R

Answer c



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Conclusion

- LLMs, especially GPT-4o, can identify and assert disjointness axioms in ontologies (relatively well).

- Prompt structure and strategy can have a significant impact on the results.

- LLMs struggle with some disjointness classifications.
  - ➢ Especially those without discrete real-world examples.

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Future Work

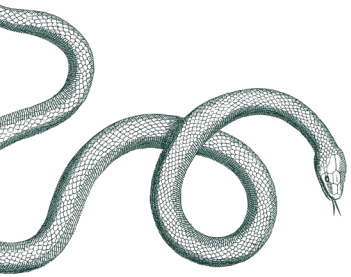
Immediate future work:

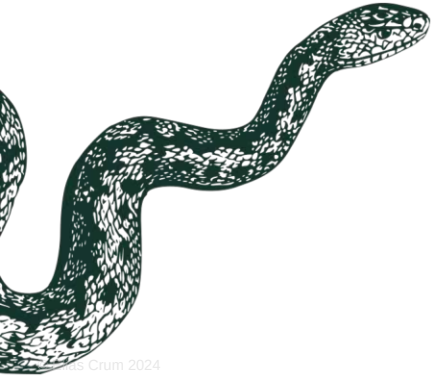
- Assess how LLM classifications are integrated with logical reasoning to minimize LLM prompts needed.
- Evaluate the proposed approach on various ontologies, LLMs and a larger dataset.
- Involve human domain experts to evaluate the results.

Further enhancements:

- Develop heuristics for selecting promising candidate pairs for disjointness.
- Explore testing classes with few non-disjoint instances to identify potential errors.
- Refine prompts with role descriptions and elements like “at the same time” or “theoretically” for improved responses.

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Thank you!



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Pairs

positive examples	negative examples
(Place, Person)	(BasketballPlayer, BaseballPlayer)
(Continent, Sea)	(Garden, HistoricPlace)
(Gene, Movie)	(President, BeautyQueen)
(Planet, Star)	(MeanOfTransportation, Reptile)
(BaseballLeague, BowlingLeague)	(Castle, Prison)

in https://DBpedia.org/ontology/

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